

## **Task 'Unit 3: Cloud Design Tools' Unit 3**

### **Formative Discussion: Pulumi as a Superior Cloud Infrastructure Design Tool**

#### **Introduction**

Pulumi is a modern Infrastructure as Code (IaC) platform that organizations can utilize to improve the definition, deployment and management of cloud infrastructure using common programming languages. Unlike Terraform, which uses HashiCorp Configuration Language (HCL) as the primary language, Pulumi supports a variety of languages for instance: Python, TypeScript, Go, C#, Java and YAML. This makes Pulumi particularly suitable for teams that already have experience in software development, programming languages in general software engineering practices, which is the basis of IaC.

(Pulumi, 2026a).

An additional advantage of Pulumi is the multi-cloud compatibility. Pulumi supports many cloud providers, for instance Amazon Web Services (AWS), Microsoft Azure and Google Cloud, enabling organizations to manage infrastructure across diverse cloud environments using a consistent IaC approach.

(Pulumi, 2026b).

For organizations adopting hybrid or multi-cloud compatibility, this could be quite useful as explained previously regarding AWS and Azure. While Terraform also provides strong multi-cloud capabilities, Pulumi's use of general-purpose programming languages which provides greater flexibility for building reusable components, applying programming abstractions and integrating infrastructure development into existing software workflows.

(Pulumi, 2026a).

## Task 'Unit 3: Cloud Design Tools' Unit 3

Pulumi can be considered superior due to another reason, that its support for testing and governance. Because infrastructure is composed of using standard programming languages, where organizations can use established testing frameworks to infrastructure code. Pulumi also provides Policy as Code, allowing security, compliance and operational requirements to be pre-defined and automatically assessed before deployment into the infrastructure.

(Pulumi, 2026c; Pulumi, 2026d).

Such factors support the governance points brought up in Unit's 2 post where organizational requirements should be integrated into cloud strategy rather than treated as secondary considerations.

In summary, whereas Terraform may be considered the more mature, mainstream IaC cloud design tool, I believe that Pulumi is superior for organizations that prioritize multi-cloud and developer flexibility, reusable infrastructure and integrated testing and governance. Its ability to treat infrastructure as software provides a strong basis for consistent and scalable cloud management.

### References

Pulumi (2026a) Pulumi vs. Terraform. Available at:

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## **Task 'Unit 3: Cloud Design Tools' Unit 3**

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Pulumi (2026c) Testing Pulumi Programs. Available at:

<https://www.pulumi.com/docs/iac/concepts/testing/> (Accessed: 6 August 2026].

Pulumi (2026d) Pulumi Policies. Available at:

<https://www.pulumi.com/docs/insights/policy-as-code/> [Accessed: 6 August 2026].

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